SIMPLE IMPACT STUDIES — SAMPLE DRAFT

Bexar County Sample

Traffic Impact Study

Prepared September 2, 2026 · 29.4640, -98.6350

Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) presents the anticipated traffic impacts of the proposed Bexar County Sample located within San Antonio-New Braunfels MSA, Texas. The study evaluates 17 intersections within a 1.00-mile study radius. The report follows the outline in TxDOT Traffic and Safety Analysis Procedures Manual (TSP) Chapter 16 Appendix Q, with capacity analysis by the openly-published Webster/Akcelik signalized control-delay method that the TSP capacity procedures are built on, and trip generation per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716). The development generates a determining peak-hour trip count of 680 vehicles, classifying it as Category 2 (500–1,000 peak-hour trips) under TSP §16.2.1 Table 16-1.

Reviewing authority: City of San Antonio.

San Antonio Development Services Department (DSD) — Land Development, per Unified Development Code §35-502 (TIA & Roughly Proportionate Determination) and UDC Appendix B §35-B122 (TIA Submittal Contents).

Findings:

• 0 intersections are projected to drop one or more LOS grades under the Background-plus-site scenario.

• 7 intersections operate at LOS E or F under the Background-plus-site scenario and are flagged for mitigation under TSP §16.4.3.

17	0	7	27.4s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

1.0 INTRODUCTION

Per TSP §16.1, this Traffic Impact Analysis determines whether the transportation infrastructure surrounding the project can accommodate the traffic demand the proposed development will introduce. It is layered with the City of San Antonio TIA standard where applicable. San Antonio Development Services Department (DSD) — Land Development, per Unified Development Code §35-502 (TIA & Roughly Proportionate Determination) and UDC Appendix B §35-B122 (TIA Submittal Contents).

TSP §16.1 also flags that this chapter does not provide final TIA recommendation thresholds — the TxDOT District and the host municipality may request analysis beyond this scope, and the project manager determines final methodology. Refer to the TxDOT Access Management Manual Chapter 3 for the TIA-request criteria that apply on the state system.

Preliminary Scoping (TSP §16.2.2)

Per TSP §16.2.2, the specific project scope should be confirmed with the local TxDOT District before the TIA begins, via an in-person meeting or alternate correspondence. Items to confirm:

• Number of access points to TxDOT roadways
• Acceptable LOS thresholds
• Selected study years
• Anticipated influence area
• Intersections and roadways to be analyzed
• Scenarios to analyze
• Data collection method
• Project schedule and buildout year
• Data source
• Use of TDM outputs, growth factors, etc.
• Other major projects in the area

Required submission: TIA + Rough Proportionality cost calculation (mitigation cost capped at the maximum proportional impact, UDC §35-502). Pre-submittal scoping meeting with TCI + Public Works + Planning is mandatory.

Host-jurisdiction TIA threshold: ≥ 76 peak-hour trips (UDC §35-502). An update-TIA is required when an increase to an existing TIA or zoning results in ≥ 76 PHT or ≥ 10% of the total PHT for the development, whichever is greater. Below 76 PHT a Peak Hour Trip Generation Form only.

2.0 PROJECT DESCRIPTION

2.1 Site Plan

Project name	Bexar County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	29.4640°, -98.6350°
Host jurisdiction	City of San Antonio

2.2 Area of Influence

The anticipated area of influence is the 1.00-mile radius surrounding the site, defined per TSP §16.3.1 to include the roads and intersections within the project and the area impacted by project-generated trips. Final area of influence is confirmed with the District during preliminary scoping.

2.3 Phasing and Timing

Single-phase buildout year: 2027. Future analysis horizon for this TIA category: Buildout year, each phase completion year, and five years past buildout (TSP §16.2.1 Category 2).

Phased development scenarios are not modeled in this screening analysis. Each phase completion year listed above should be analyzed separately in the formal submittal — phase boundaries and completion dates must come from the applicant's construction schedule.

3.0 STUDY AREA CONDITIONS

3.1 Existing and Anticipated Land Use

Existing land use and the host-jurisdiction Future Land Use Plan for parcels within the area of influence should be compiled from the host city's adopted Comprehensive Plan and any applicable overlay districts. This screening report does not generate that inventory [placeholder — requires GIS pull against host-jurisdiction parcel layer].

3.2 Existing and Future Roadway System

Roadway functional class, lane count, posted speed, and surface for state-system routes (IH / US / SH / FM / RM / BU / BS / SL / SS) are referenced against TxDOT Roadway Inventory (RHiNo) and the TxDOT Statewide Planning Map. Future roadway system context is taken from the TxDOT Unified Transportation Program (UTP) and the regional MPO Metropolitan Transportation Plan (MTP) where in-area projects are programmed.

4.0 EXISTING OPERATIONS

4.1 Roadway Conditions and Traffic Controls

Geometry, signal timing, and traffic-control inventory for the study network are derived from RHiNo plus host-jurisdiction signal-timing records. For the formal submittal, supplementary field inspection within 12 months of submittal is recommended.

4.2 Alternate Modes

Pedestrian, bicycle, and transit facility inventories within the area of influence should be confirmed against host-jurisdiction GIS and regional transit-operator maps. Per TSP §16.3.3, multimodal reduction is applied to trip generation in areas where alternate transit is readily available — this screening report does not auto-apply multimodal reduction.

4.3 Traffic Volumes

Existing AADT for state-system segments is taken from the TxDOT Open Data Portal AADT layer (annual refresh). Peak-hour turning-movement counts at study intersections should be collected mid-week (Tue/Wed/Thu), school-in-session, within 12 months of submittal — per Houston IDM

§15.06.01.A counts must be within 12 months in high-growth areas and within 24 months elsewhere; Austin TDS no longer accepts pre-COVID counts by default.

4.4 Level of Service

Affected intersection	Distance (mi)	Existing LOS	Existing delay (s)
Ingram Road & Potranco Road	0.21	B	14.8
Ingram Road & Culebra Road	0.38	B	18.5
Potranco Road & Culebra Road	0.61	B	18.0
Ingram Road & North Western	0.61	C	25.7
Reed Road & Culebra Road	0.61	F	300.0
Culebra Road & Pipers Lane	0.73	F	300.0
Northwest Loop 410 & Culebra Road	0.78	C	22.4
Ingram Road & Wurzbach Road	0.80	F	139.3
Culebra Road & Timber View Drive	0.82	F	300.0
Wurzbach Road & Timberhill	0.82	C	20.4
Culebra Road & Connally Loop	0.83	F	300.0
Micron Drive & Reed Road	0.84	B	15.5
Potranco Road & Micron Drive	0.91	D	41.6
Ingram Road & Northwest Loop 410	0.92	F	139.3
Wurzbach Road & Crystal Hill	0.94	B	14.1
Culebra Road & Rim Rock Trail	0.98	F	300.0
Ingram Road & Connally Loop	0.99	D	36.1

4.5 Safety

Crash history for the most recent five years on study segments and intersections should be pulled from the TxDOT Crash Records Information System (CRIS Public Query). This screening report does not auto-generate the crash summary [placeholder — requires CRIS pull].

5.0 PROJECTED TRAFFIC

5.1 Site Generated Traffic

Trip generation is calculated per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at a proposed size of 85 1,000 sqft GLA. Per TSP §16.3.3, the fitted curve equation is preferred when the data plot contains at least 20 data points or has an R^2 of at least 0.75; otherwise average rates apply. Daily, AM peak, and PM peak hour trips are reported below.

Period	Entering trips	Exiting trips
Daily	3,400	3,400

Period	Entering trips	Exiting trips
AM peak hour	166	106
PM peak hour	326	354

Internal capture (per TSP §16.3.3: trips between land uses within the same development that do not touch the off-site street system) and pass-by trips (already traveling on the adjacent roadway network and entering the proposed development) are accounted for below.

Pass-by capture applied	30%
Internal capture applied	0%

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

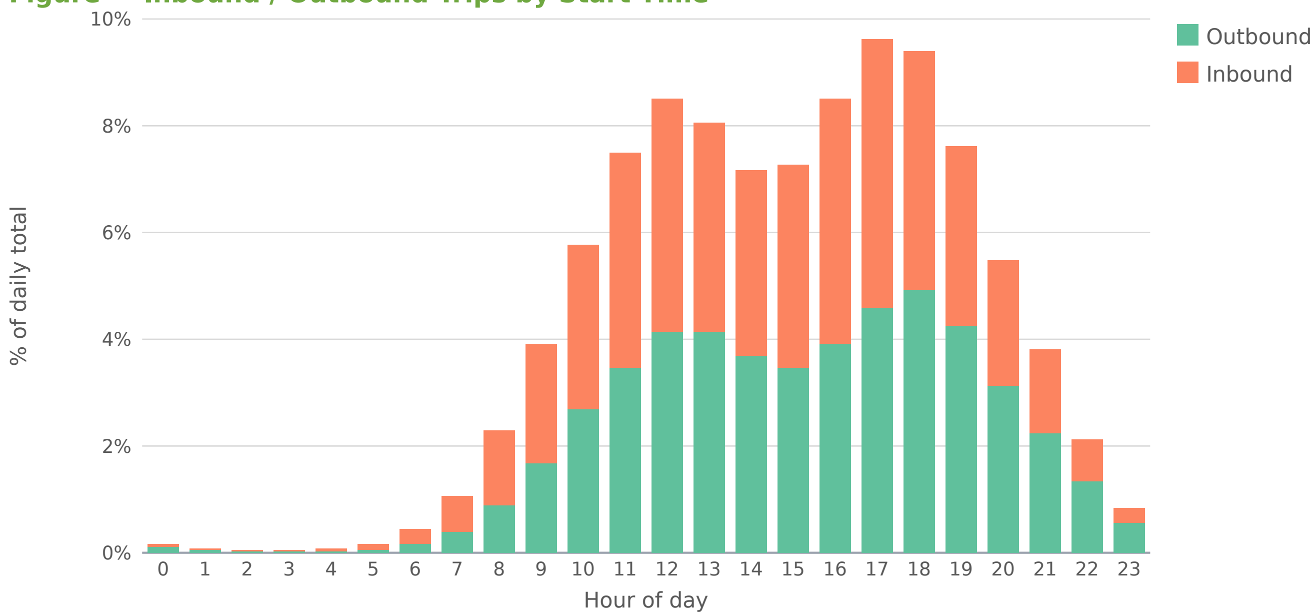
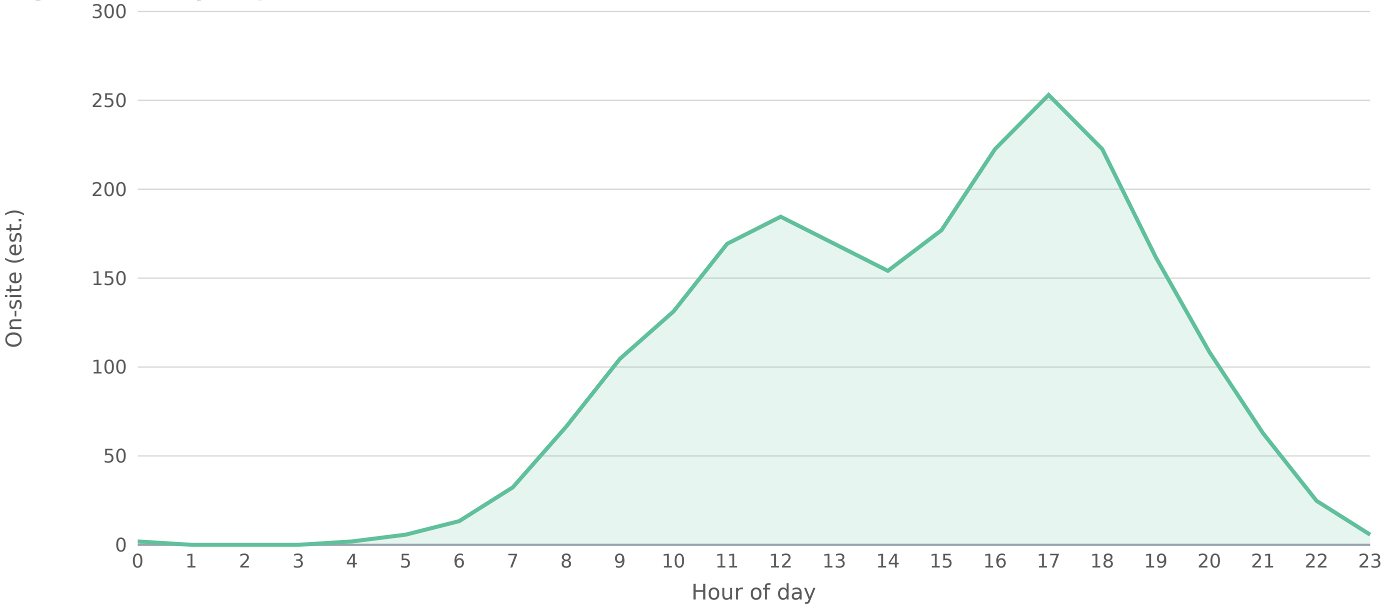


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

5.2 Trip Distribution

Per TSP §16.4, project distribution is assigned using engineering judgment, informed by the surrounding roadway network geometry and proximity to the project access points. If only one project driveway is proposed, all trips enter and exit through that driveway. Final distribution percentages are confirmed with the District during preliminary scoping.

5.3 Trip Assignment

Site-generated external trips are assigned to the study network using inverse-distance weighting from the project site to each signalized intersection, normalized so the period total matches the external-trip count from §5.1.

5.4 Non-Site Traffic

Background traffic is grown at 1.5% per year over 1 year. Per TSP §16.4.2, the prescribed method is to average at least the last five years of historical AADT data for the segment analyzed to derive an average annual growth rate; the value applied here is a screening default and should be re-calibrated to the affected segments' five-year AADT trend before formal submittal. Background growth data is also commonly sourced from the host city or regional MPO travel-demand model (H-GAC, NCTCOG, CAMPO, or AAMPO depending on the MSA).

Other major projects in the area (committed developments) must be coordinated with the District and governing municipality per TSP §16.3.3 and added on top of the AADT-derived background growth. This screening analysis does not auto-pull committed-development trips.

5.5 Total Traffic

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	227	138	89
PM Peak Hour	680	204	0	429	206	223
Saturday Midday	748	56	0	624	312	312
Daily Total	6,800	510	0	5,671	2,836	2,835

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross − pass-by − internal-capture) × auto-mode share (90.1% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

5.6 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	27%
Southeast (ESE + SSE)	19%
Southwest (SSW + WSW)	9%
Northwest (WNW + NNW)	45%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Culebra Road & Connally Loop	SSE	0.83	3,895	0.1	12.31%
Reed Road & Culebra Road	WNW	0.61	3,447	0.1	11.80%
Culebra Road & Pipers Lane	WNW	0.73	3,447	0.1	11.32%
Culebra Road & Timber View Drive	WNW	0.82	3,447	0.1	11.01%
Culebra Road & Rim Rock Trail	WNW	0.98	3,447	0.1	10.42%
Ingram Road & Wurzbach Road	ENE	0.80	2,216	0.1	7.18%
Ingram Road & Northwest Loop 410	ENE	0.92	2,216	0.1	6.91%
Potranco Road & Micron Drive	SSW	0.91	1,669	0.1	5.22%
Ingram Road & Connally Loop	ENE	0.99	1,592	0.0	4.86%
Ingram Road & North Western	ENE	0.61	1,298	0.0	4.48%
Northwest Loop 410 & Culebra Road	SSE	0.78	1,117	0.0	3.64%
Wurzbach Road & Timberhill	ENE	0.82	956	0.0	3.08%
Ingram Road & Culebra Road	SSW	0.38	773	0.0	2.88%
Potranco Road & Culebra Road	SSE	0.61	706	0.0	2.44%
Micron Drive & Reed Road	WSW	0.84	361	0.0	1.16%
Ingram Road & Potranco Road	SSE	0.21	241	0.0	0.95%
Wurzbach Road & Crystal Hill	ENE	0.94	100	0.0	0.31%

17 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

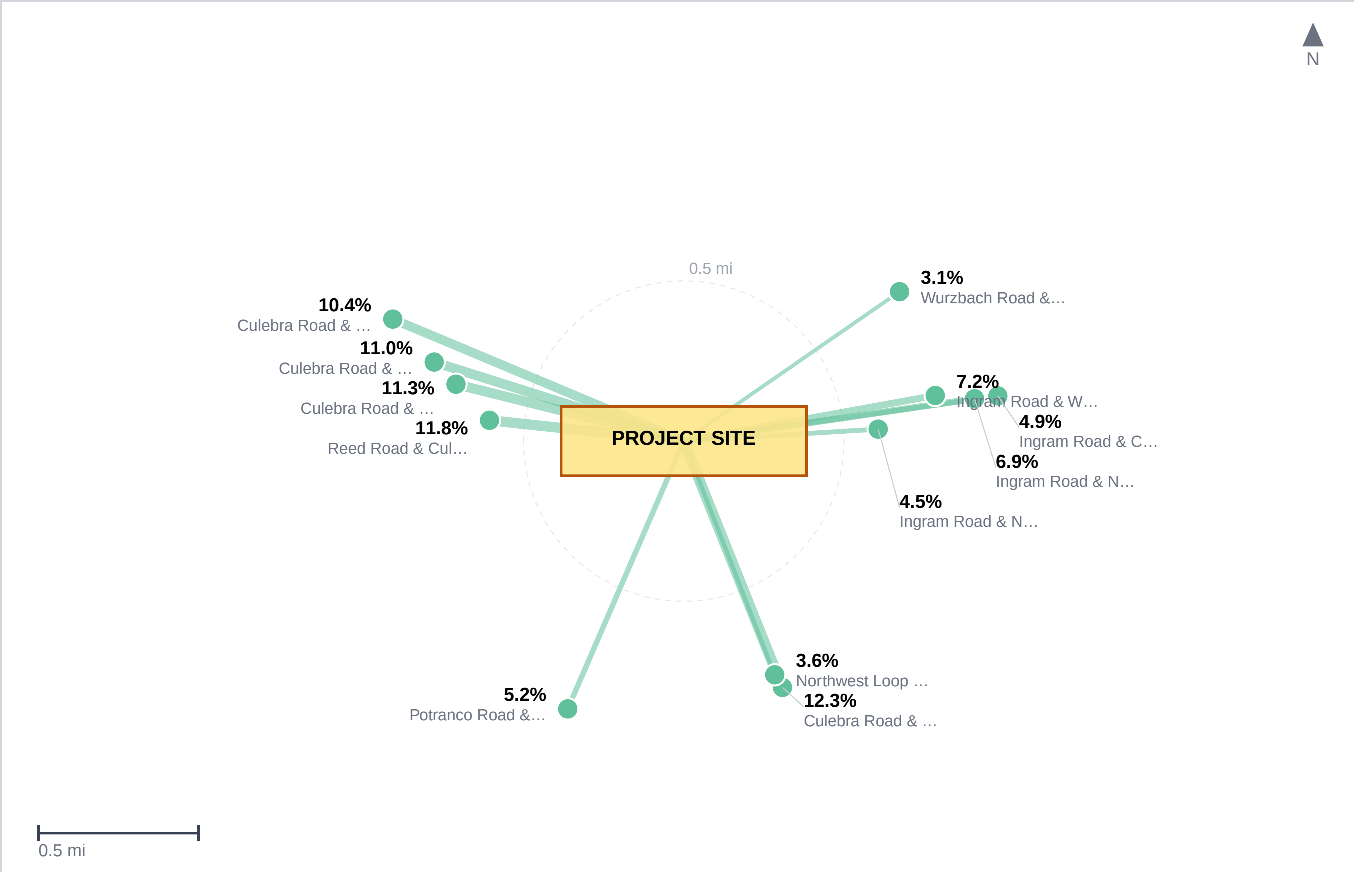
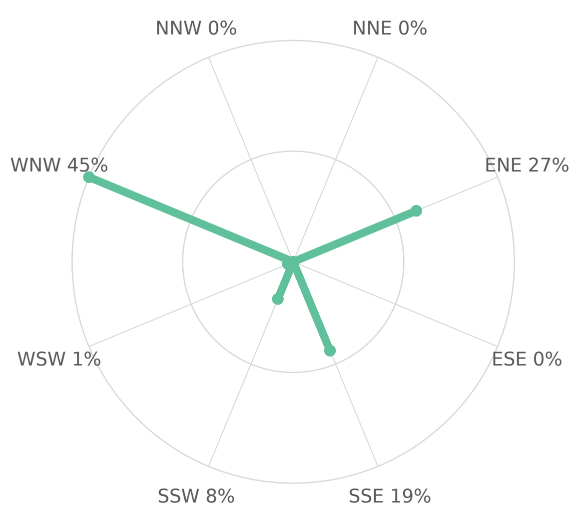


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

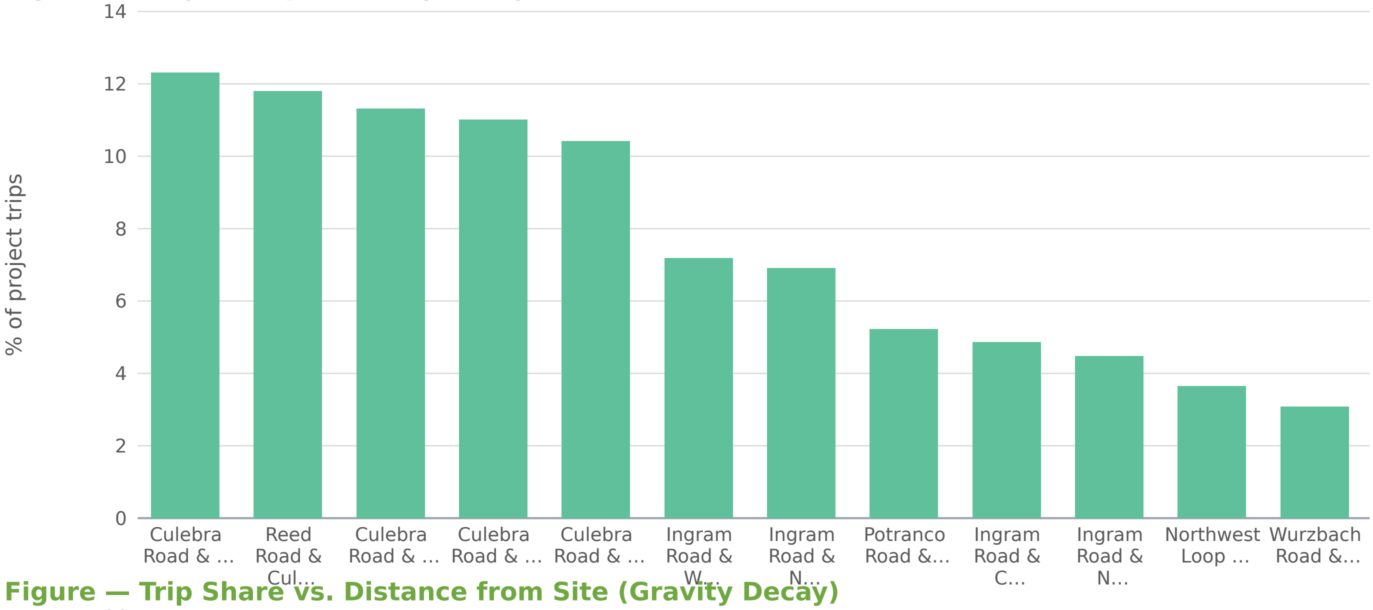


Figure — Trip Share vs. Distance from Site (Gravity Decay)

6.0 TRAFFIC AND IMPROVEMENT ANALYSIS

6.1 Site Access

Where the project fronts a state-system roadway (IH / US / SH / FM / RM / BU / BS / SL / SS), site access is reviewed through the TxDOT Driveway Access Permit (DAP) process under the Access Management Manual Chapter 3 §3. Driveway spacing, geometry, and auxiliary lanes (right-turn deceleration, left-turn) are checked against ACM Chapter 2 §3 (Table 2-2 spacing) and the Roadway Design Manual Chapter 16.

6.2 Auxiliary Lane and Sight Distance Analysis

Auxiliary lane warrants (right-turn deceleration, left-turn) per ACM Chapter 2 §4 and Roadway Design Manual Chapter 16 should be checked against the project's access geometry. Intersection sight distance per RDW Ch. 2 should be verified at every proposed driveway. This screening report does not perform per-driveway sight-distance or auxiliary-lane warrant calculations [placeholder — requires final driveway geometry].

6.3 Capacity and Level of Service Analysis

Per TSP §16.4.1 and §16.4.4, submittal capacity analysis follows the latest HCM methodology using one of the District-accepted tools (Synchro, HCS, Vissim, or Vistro); this screening applies the openly-published Webster/Akçelik signalized model pending that tool run. For signalized intersections, each approach and the overall intersection are analyzed. Two future scenarios are compared at each affected intersection: Background (grown traffic without the proposed project) and Background-plus-site (grown traffic plus the project's external trips at the assigned distribution). An Existing scenario is included for context. Host-jurisdiction LOS standard: LOS D generally; LOS E inside Transit-Oriented Development overlays per UDC §35-208.

Intersection	Existing LOS	Background LOS	Bgd+Site LOS	Δ delay (s)	Q95 (ft)
Ingram Road & Potranco Road	B	B	B	0.9	88
Ingram Road & Culebra Road	B	B	B	0.2	157
Potranco Road & Culebra Road	B	B	B	0.2	142
Ingram Road & North Western	C	C	C	3.1	312
Reed Road & Culebra Road	F	F	F	0.0	1,049
Culebra Road & Pipers Lane	F	F	F	0.0	1,082
Northwest Loop 410 & Culebra Road	C	C	C	0.1	231
Ingram Road & Wurzbach Road	F	F	F	27.4	827
Culebra Road & Timber View Drive	F	F	F	0.0	1,124
Wurzbach Road & Timberhill	C	C	C	0.1	179
Culebra Road & Connally Loop	F	F	F	0.0	1,208
Micron Drive & Reed Road	B	B	B	0.0	60
Potranco Road & Micron Drive	D	D	D	3.6	413
Ingram Road & Northwest Loop 410	F	F	F	14.2	601
Wurzbach Road & Crystal Hill	B	B	B	0.0	17
Culebra Road & Rim Rock Trail	F	F	F	0.0	1,270
Ingram Road & Connally Loop	D	D	D	2.1	390

6.4 Mitigation

Per TSP §16.4.3, any operational deficiencies found in the future analysis are considered for mitigation. The threshold for acceptable operations (LOS, queue length, travel time, and other MOEs) is agreed upon with the District during preliminary scoping rather than being set as a single statewide standard. Typical mitigation measures listed in TSP §16.4.3 include: right-turn deceleration lanes, left-turn lanes, median modifications, traffic signal modification and installation, road widening, revised striping, turning lane restrictions, and alternative intersections / interchanges. The developer is responsible for implementing the agreed-upon mitigation measures.

Screening Mitigation Recommendations

Reed Road & Culebra Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Culebra Road & Pipers Lane [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Ingram Road & Wurzbach Road [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site

driveway alignment or development scale if delay remains above 80s.

Culebra Road & Timber View Drive [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Culebra Road & Connally Loop [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Potranco Road & Micron Drive [MINOR]
Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Ingram Road & Northwest Loop 410 [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Culebra Road & Rim Rock Trail [MAJOR]
Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Ingram Road & Connally Loop [MINOR]
Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Rough Proportionality Cap (UDC §35-502)

Per UDC §35-502, if the proposed mitigation cost is less than or roughly equal to the maximum proportional impact, the mitigation is considered roughly proportionate; if it exceeds that maximum, the City must limit required mitigation to an amount roughly equal to the maximum proportional impact. A Rough Proportionality cost calculation must be prepared and submitted with this TIA; this screening report does not generate that calculation [placeholder — requires final mitigation cost estimate and the City's proportionality worksheet].

6.5 Driveway Operational Analysis

Per-driveway operational analysis (full-movement vs. left-in/left-out configuration, throat depth, on-site queue spillback) depends on the final site plan and is not included in this screening-level TIA [placeholder].

7.0 SAFETY ANALYSIS

Per the Appendix Q outline, Safety Analysis is a standalone section separate from §4.5 Existing Operations — Safety. It includes the project's effect on study-area crash trends, sight-distance impacts at proposed access, and any HSIP-identified crash clusters within the area of influence. This screening report does not auto-generate the safety analysis [placeholder — requires CRIS pull + sight-distance verification].

8.0 CONCLUSIONS

Based on the screening analysis above, the project is classified as Category 2 (500–1,000 peak-hour trips) per TSP §16.2.1. Of the 17 affected intersections analyzed, 0 drop one or more LOS grades and 7 operate at LOS E or F under the Background-plus-site scenario. The horizon analyzed in this screening is the buildout year only; the full submittal must cover the years required by the project's TIA category per TSP §16.2.1 Table 16-1.

9.0 RECOMMENDATIONS

Submit this TIA to City of San Antonio, with parallel TxDOT-District coordination where any state-system route is in frontage. Validate the screening results against current-edition manuals, updated turning-movement counts within the most recent 12 months (Houston IDM §15.06.01.A: 12 months in high-growth areas, 24 months elsewhere), and the preliminary-scoping outcome with the District. The report must be sealed by a Texas-licensed Professional Engineer per 22 TAC §137.33 (Sealing Procedures), promulgated under The Texas Engineering Practice Act (Tex. Occ. Code Ch. 1001), with the seal on the cover and on every sealed sheet.

10.0 APPENDICES

The formal submittal appendices, per the Appendix Q outline, should include: scoping correspondence with the TxDOT District; site plan figures; TMC count sheets with date, weather, observer; Trip-generation worksheets with rate/equation selection rationale; HCS / Synchro / Vissim / Vistro output; signal warrant analyses citing the TMUTCD (2025 edition, effective Jan 18, 2026); auxiliary-lane and sight-distance worksheets; CRIS crash data summary; and the host-jurisdiction's submission forms (Houston Access Management Data Summary Form + CPC 101 Form / Austin TIA Determination + Scope / Dallas TIS Waiver / Fort Worth TIA Worksheet / San Antonio TIA Threshold Worksheet + Rough Proportionality calculation).

Programmed Projects (informational)

Review of programmed transportation projects within the area of influence should consult the TxDOT Unified Transportation Program (UTP 2026, adopted Aug 2025), the federally-required Statewide Transportation Improvement Program (STIP), and Alamo Area MPO (AAMPO) TIP. This screening analysis does not automatically integrate programmed-projects data; manual review is recommended for any submittal.

FINDINGS

• Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).

• Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).

• Auto-mode share of 90% applied to external trips for San Antonio-New Braunfels MSA; 10% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.

• Existing volumes were grown by 1.50%/yr over 1 year (×1.01) to the opening-year horizon.

• 17 signalized intersections fall within the study area; 0 are projected to drop at least one LOS grade after build-out.

• 7 intersections are projected to operate at LOS E or F under the build condition and require formal mitigation per San Antonio Public Works Department TIS guidance.

• Worst-impact location: Ingram Road & Wurzbach Road — projected delay rises 27.4s (LOS F → F); 95th-pct queue 827 ft on the critical approach.

• Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a ±0.5%/yr background growth band around the applied value.

METHODOLOGY NOTES

• Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.

• Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.

• Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.

• Weather adjustment applies published rain/snow capacity-reduction factors as adopted in US traffic-engineering practice: clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.

• Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: a production-constrained gravity model $T_j = P \cdot (A_j \cdot F_j) / \Sigma(A_x \cdot F_x)$ allocates trips to surrounding signals, where attractiveness A_j is the signal's through-volume and the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work coefficients a=28507, b=-0.02, c=-0.123) on the travel time t to each signal. Step 3 Mode Choice: a binary logit $P(auto)=1/(1+e^{-(ASC-\lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.

• After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.

• Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.

• Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.

• Intersection-level control delay uses the standard signalized-intersection model $d = d1 + d2$ (Webster uniform delay + Akçelik time-dependent incremental-delay term, both openly published) with a 90s cycle, $g/C = 0.45$, 1,800 vphpl saturation flow (× weather factor), 15-minute peak analysis period ($T = 0.25$ hr) and pretimed-signal incremental-delay factor $k = 0.5$. Reported control delay is capped at 300 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

• Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal allocation perturbed ±15% from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (standard cyclic-queue relation with Poisson percentile factor: $Q95 \approx Q1 \times 1.65 \times 25$ ft/veh) are reported.

• Level of Service is assigned from the conventional US signalized control-delay bands republished in state DOT traffic-engineering manuals: A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s.

• Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a ±0.5%/yr growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

• Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

• Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as (turbo lanes / approach lanes) × (1 − g/C) / (g/C), with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%…+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters b=-0.02, c=-0.123 (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Reed Road & Culebra Road	0.61	1.5	278	21.5%
Ingram Road & Potranco Road	0.21	0.5	151	11.7%
Culebra Road & Pipers Lane	0.73	1.8	145	11.2%
Culebra Road & Timber View Drive	0.82	2.0	145	11.2%
Culebra Road & Rim Rock Trail	0.98	2.4	145	11.2%
Ingram Road & North Western	0.61	1.5	116	9.0%
Ingram Road & Wurzbach Road	0.80	1.9	116	9.0%
Ingram Road & Northwest Loop 410	0.92	2.2	60	4.6%
Potranco Road & Micron Drive	0.91	2.2	36	2.8%
Ingram Road & Connally Loop	0.99	2.4	30	2.3%
Potranco Road & Culebra Road	0.61	1.5	21	1.6%
Ingram Road & Culebra Road	0.38	0.9	19	1.5%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 90% applied via a binary logit $P(auto) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 10% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c: 0.27. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	2.97	233	0.27
Minor arterial	3.07	196	0.20
Principal arterial	0.55	75	0.19

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 7 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 236 network junctions, max imbalance 0.00 veh. 11 study intersection(s) carry path-derived movements; 6 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

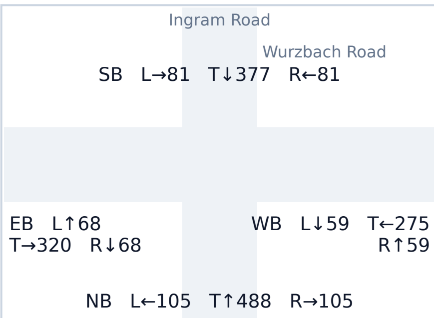
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Ingram Road & Wurzbach Road

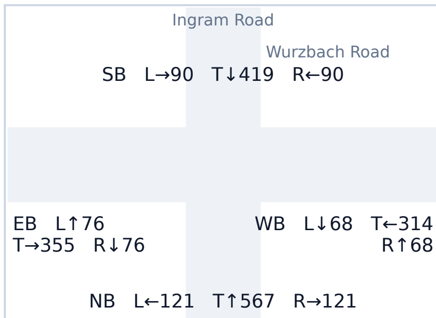
Signal ID	san-antonio-1542
Location	SW San Antonio-New Braunfels · 0.80 mi from site
Added PM peak trips	116
Existing / No-Build (PM)	LOS F · 147.1 s/veh · v/c 1.25
Build (PM)	LOS F · 174.5 s/veh · v/c 1.31
Δ control delay	+27.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

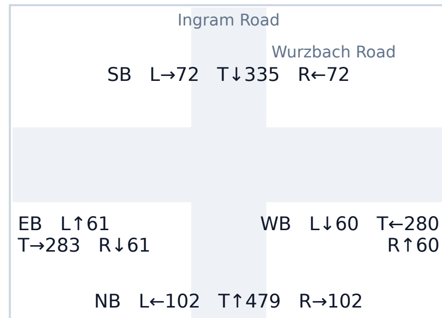
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	749	60	809	0.92 → 1.00	41.1 → 55.8	D → E	827
SB	599	0	599	0.74 → 0.74	26.4 → 26.4	C → C	509
EB	507	0	507	0.63 → 0.63	22.6 → 22.6	C → C	400
WB	395	56	450	0.49 → 0.56	19.5 → 20.9	B → C	341

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Right	60	52%
WB Left	56	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

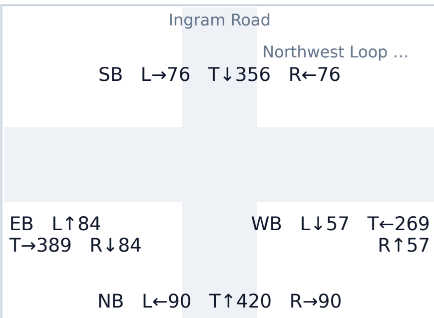
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Ingram Road & Northwest Loop 410

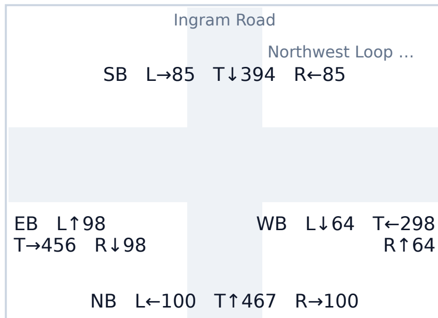
Signal ID	san-antonio-1877
Location	SW San Antonio-New Braunfels · 0.92 mi from site
Added PM peak trips	60
Existing / No-Build (PM)	LOS F · 147.1 s/veh · v/c 1.25
Build (PM)	LOS F · 161.3 s/veh · v/c 1.28
Δ control delay	+14.2 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

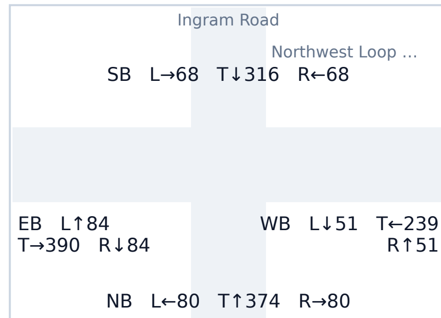
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	667	0	667	0.82 → 0.82	30.9 → 30.9	C → C	601
SB	564	0	564	0.70 → 0.70	24.8 → 24.8	C → C	466
EB	592	60	652	0.73 → 0.81	26.1 → 29.7	C → C	580
WB	426	0	426	0.53 → 0.53	20.3 → 20.3	C → C	316

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	60	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

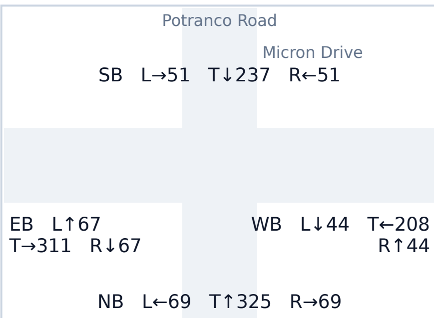
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Potranco Road & Micron Drive

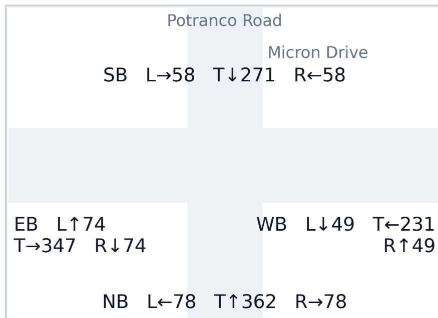
Signal ID	san-antonio-2924
Location	SW San Antonio-New Braunfels · 0.91 mi from site
Added PM peak trips	36
Existing / No-Build (PM)	LOS D · 43.8 s/veh · v/c 0.94
Build (PM)	LOS D · 47.4 s/veh · v/c 0.96
Δ control delay	+3.6 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

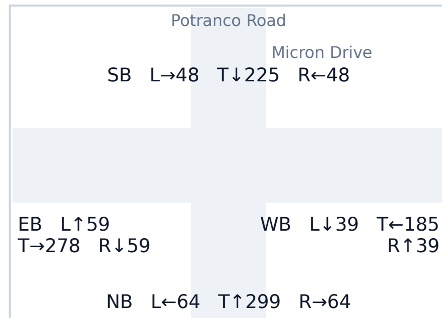
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	501	17	518	0.62 → 0.64	22.4 → 23.0	C → C	413
SB	369	19	387	0.46 → 0.48	19.0 → 19.4	B → B	280
EB	495	0	495	0.61 → 0.61	22.2 → 22.2	C → C	387
WB	329	0	329	0.41 → 0.41	18.2 → 18.2	B → B	229

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	19	53%
NB Through	17	47%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

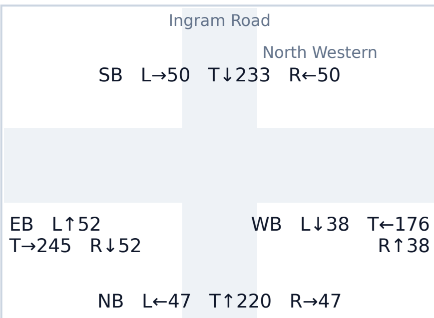
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Ingram Road & North Western

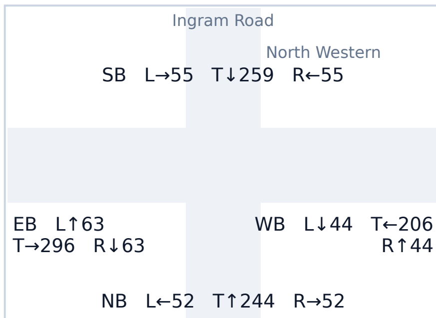
Signal ID	san-antonio-1262
Location	SW San Antonio-New Braunfels · 0.61 mi from site
Added PM peak trips	116
Existing / No-Build (PM)	LOS C · 26.1 s/veh · v/c 0.73
Build (PM)	LOS C · 29.2 s/veh · v/c 0.80
Δ control delay	+3.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

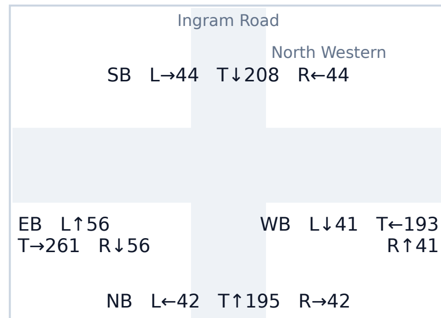
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	348	0	348	0.43 → 0.43	18.5 → 18.5	B → B	245
SB	369	0	369	0.46 → 0.46	19.0 → 19.0	B → B	264
EB	362	60	422	0.45 → 0.52	18.8 → 20.2	B → C	312
WB	238	56	294	0.29 → 0.36	16.6 → 17.5	B → B	199

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	60	52%
WB Through	56	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

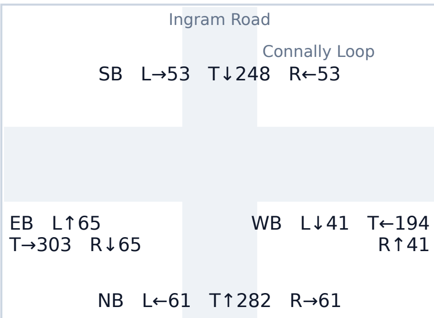
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Ingram Road & Connally Loop

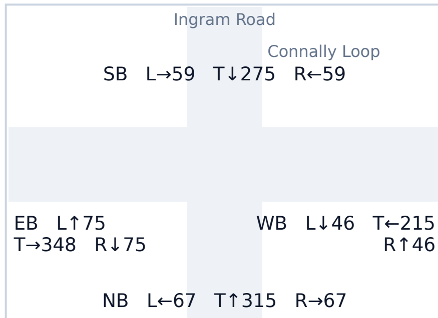
Signal ID	san-antonio-1878
Location	SW San Antonio-New Braunfels · 0.99 mi from site
Added PM peak trips	30
Existing / No-Build (PM)	LOS D · 37.6 s/veh · v/c 0.90
Build (PM)	LOS D · 39.7 s/veh · v/c 0.91
Δ control delay	+2.1 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)

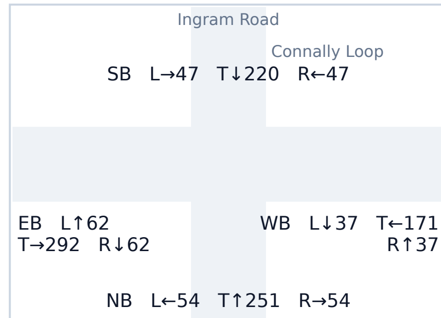
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	449	0	449	0.55 → 0.55	20.9 → 20.9	C → C	339
SB	393	0	393	0.49 → 0.49	19.5 → 19.5	B → B	285
EB	468	30	498	0.58 → 0.61	21.4 → 22.3	C → C	390
WB	307	0	307	0.38 → 0.38	17.8 → 17.8	B → B	210

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	30	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

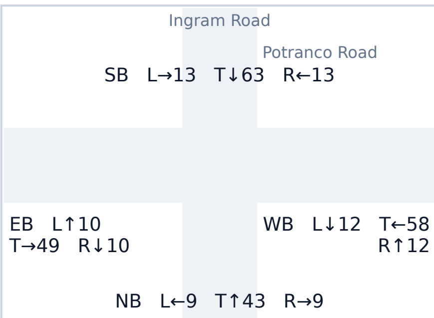
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Ingram Road & Potranco Road

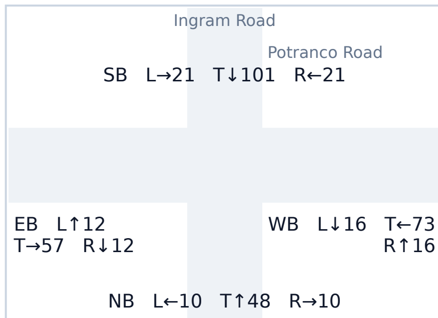
Signal ID	san-antonio-2304
Location	SW San Antonio-New Braunfels · 0.21 mi from site
Added PM peak trips	151
Existing / No-Build (PM)	LOS B · 14.8 s/veh · v/c 0.14
Build (PM)	LOS B · 15.7 s/veh · v/c 0.22
Δ control delay	+0.9 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

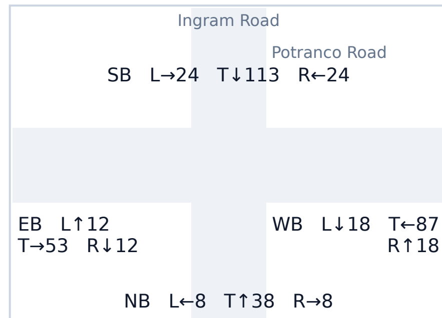
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	68	0	68	0.08 → 0.08	14.3 → 14.3	B → B	40
SB	64	79	143	0.08 → 0.18	14.3 → 15.3	B → B	88
EB	64	17	81	0.08 → 0.10	14.3 → 14.5	B → B	48
WB	49	55	105	0.06 → 0.13	14.1 → 14.8	B → B	63

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Left	60	40%
WB Right	55	36%
SB Right	19	13%
EB Left	17	11%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

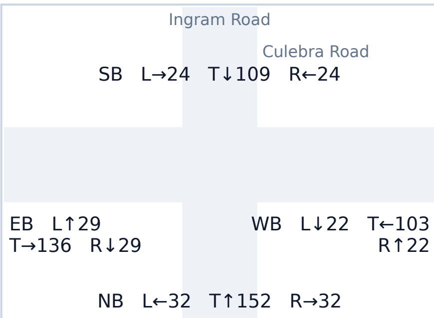
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Ingram Road & Culebra Road

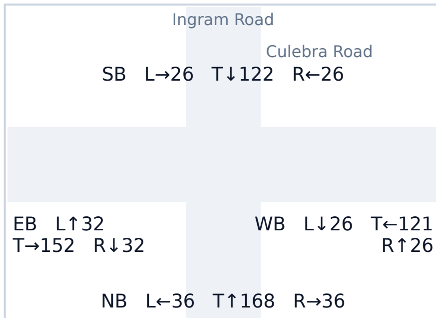
Signal ID	san-antonio-2788
Location	SW San Antonio-New Braunfels · 0.38 mi from site
Added PM peak trips	19
Existing / No-Build (PM)	LOS B · 18.6 s/veh · v/c 0.44
Build (PM)	LOS B · 18.8 s/veh · v/c 0.45
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

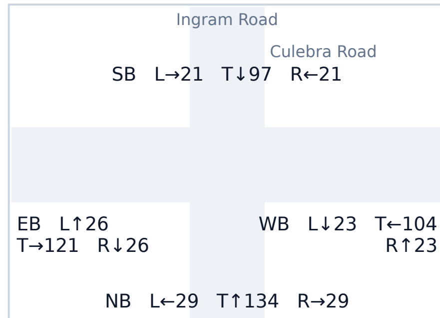
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	240	0	240	0.30 → 0.30	16.6 → 16.6	B → B	157
SB	174	0	174	0.21 → 0.21	15.7 → 15.7	B → B	109
EB	216	0	216	0.27 → 0.27	16.3 → 16.3	B → B	139
WB	155	19	173	0.19 → 0.21	15.4 → 15.7	B → B	109

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	19	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

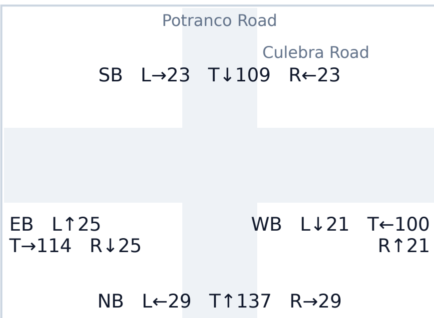
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Potranco Road & Culebra Road

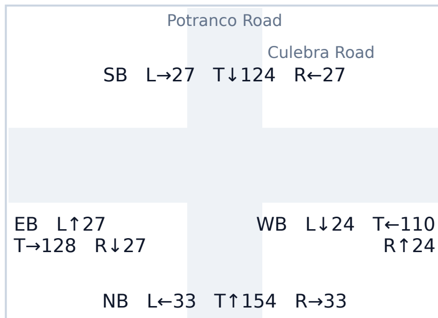
Signal ID	san-antonio-2303
Location	SW San Antonio-New Braunfels · 0.61 mi from site
Added PM peak trips	21
Existing / No-Build (PM)	LOS B · 18.0 s/veh · v/c 0.40
Build (PM)	LOS B · 18.2 s/veh · v/c 0.41
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

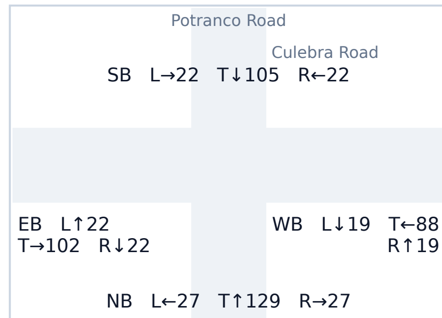
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	210	10	220	0.26 → 0.27	16.2 → 16.3	B → B	142
SB	167	11	178	0.21 → 0.22	15.6 → 15.7	B → B	112
EB	182	0	182	0.22 → 0.22	15.8 → 15.8	B → B	115
WB	158	0	158	0.19 → 0.19	15.5 → 15.5	B → B	98

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	11	52%
NB Through	10	48%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

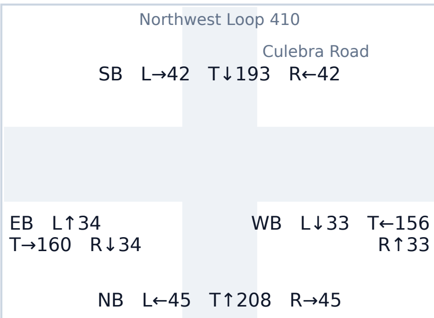
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Northwest Loop 410 & Culebra Road

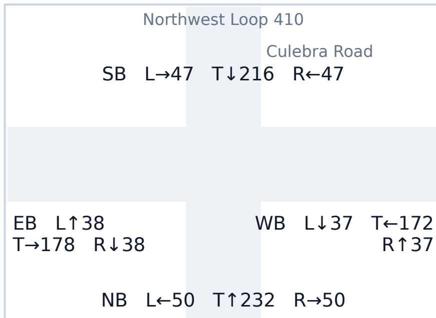
Signal ID	san-antonio-2299
Location	SW San Antonio-New Braunfels · 0.78 mi from site
Added PM peak trips	9
Existing / No-Build (PM)	LOS C · 22.7 s/veh · v/c 0.63
Build (PM)	LOS C · 22.8 s/veh · v/c 0.63
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

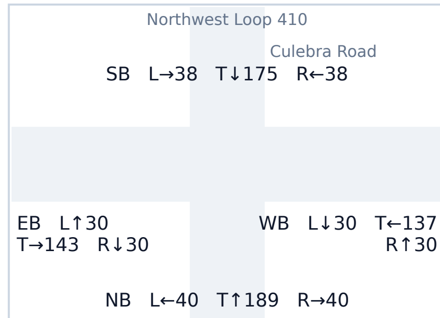
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	328	4	332	0.41 → 0.41	18.2 → 18.2	B → B	231
SB	306	5	310	0.38 → 0.38	17.7 → 17.8	B → B	213
EB	254	0	254	0.31 → 0.31	16.9 → 16.9	B → B	167
WB	246	0	246	0.30 → 0.30	16.7 → 16.7	B → B	162

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	5	56%
NB Through	4	44%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

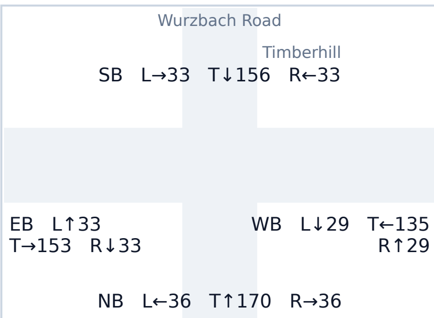
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Wurzbach Road & Timberhill

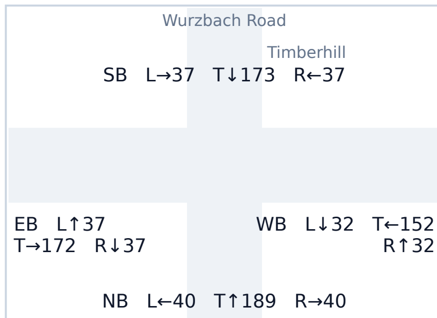
Signal ID	san-antonio-523
Location	SW San Antonio-New Braunfels · 0.82 mi from site
Added PM peak trips	7
Existing / No-Build (PM)	LOS C · 20.5 s/veh · v/c 0.54
Build (PM)	LOS C · 20.6 s/veh · v/c 0.54
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

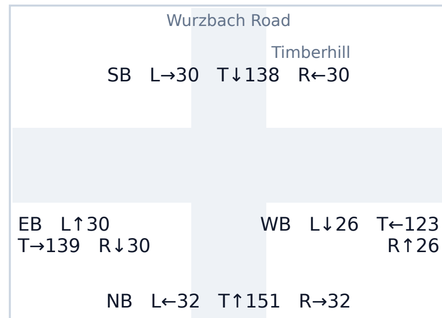
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	269	0	269	0.33 → 0.33	17.1 → 17.1	B → B	179
SB	247	0	247	0.30 → 0.30	16.8 → 16.8	B → B	162
EB	242	4	246	0.30 → 0.30	16.7 → 16.7	B → B	162
WB	212	3	216	0.26 → 0.27	16.2 → 16.3	B → B	139

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	4	57%
WB Through	3	43%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

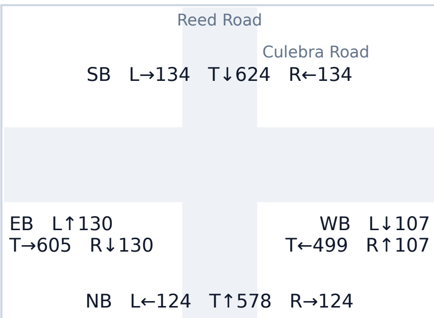
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Reed Road & Culebra Road

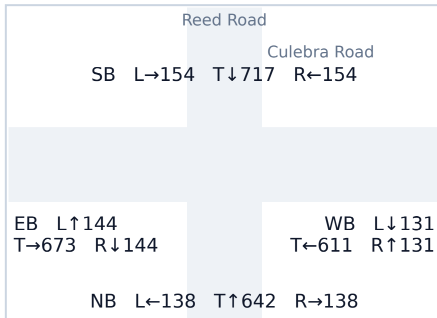
Signal ID	san-antonio-69
Location	SW San Antonio-New Braunfels · 0.61 mi from site
Added PM peak trips	278
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.94
Build (PM)	LOS F · 300.0 s/veh · v/c 2.10
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

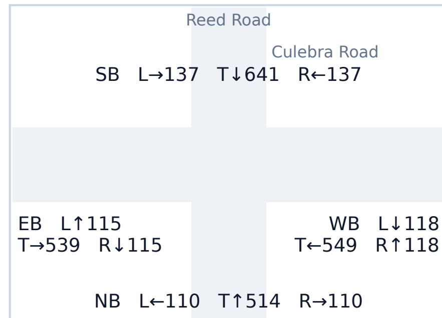
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	918	0	918	1.13 → 1.13	99.4 → 99.4	F → F	939
SB	892	133	1,025	1.10 → 1.27	87.4 → 153.8	F → F	1,049
EB	961	0	961	1.19 → 1.19	121.0 → 121.0	F → F	983
WB	728	145	873	0.90 → 1.08	37.7 → 79.1	D → E	893

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	145	52%
SB Through	133	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

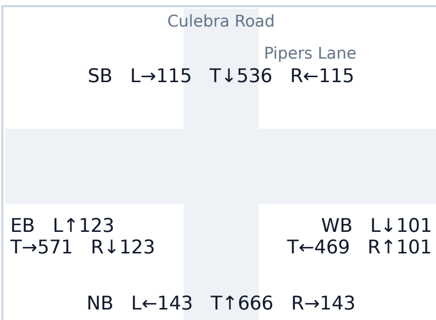
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Culebra Road & Pipers Lane

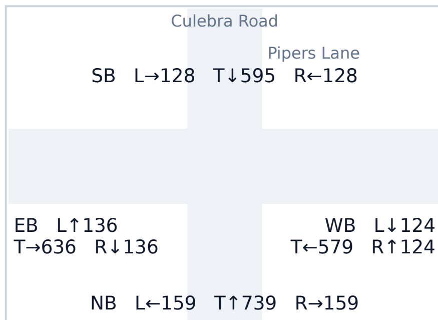
Signal ID	san-antonio-3148
Location	SW San Antonio-New Braunfels · 0.73 mi from site
Added PM peak trips	145
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.94
Build (PM)	LOS F · 300.0 s/veh · v/c 2.02
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

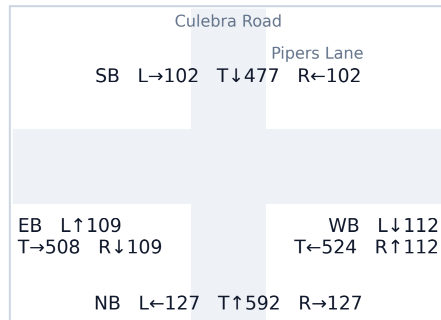
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,057	0	1,057	1.31 → 1.31	170.9 → 170.9	F → F	1,082
SB	851	0	851	1.05 → 1.05	70.4 → 70.4	E → E	871
EB	908	0	908	1.12 → 1.12	94.8 → 94.8	F → F	929
WB	682	145	827	0.84 → 1.02	32.3 → 61.5	C → E	846

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	145	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

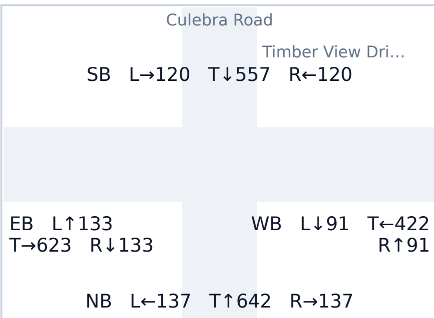
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Culebra Road & Timber View Drive

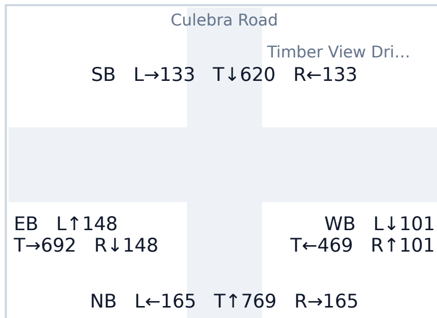
Signal ID	san-antonio-3132
Location	SW San Antonio-New Braunfels · 0.82 mi from site
Added PM peak trips	145
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.94
Build (PM)	LOS F · 300.0 s/veh · v/c 2.02
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

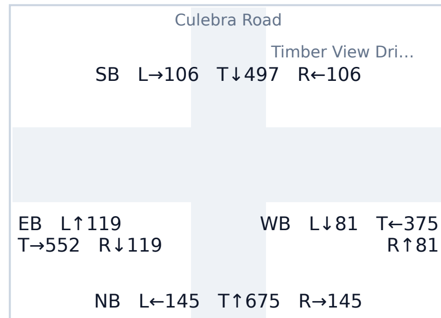
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	954	145	1,099	1.18 → 1.36	117.3 → 192.9	F → F	1,124
SB	886	0	886	1.09 → 1.09	84.8 → 84.8	F → F	906
EB	988	0	988	1.22 → 1.22	134.4 → 134.4	F → F	1,010
WB	671	0	671	0.83 → 0.83	31.3 → 31.3	C → C	607

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Left	145	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

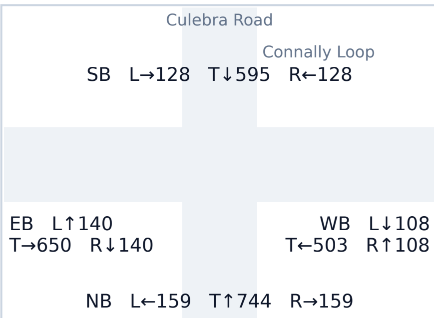
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.14 Culebra Road & Connally Loop

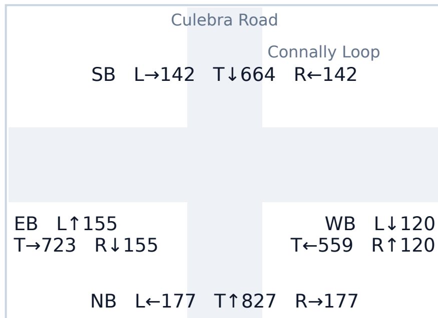
Signal ID	san-antonio-2298
Location	SW San Antonio-New Braunfels · 0.83 mi from site
Added PM peak trips	7
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 2.20
Build (PM)	LOS F · 300.0 s/veh · v/c 2.20
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

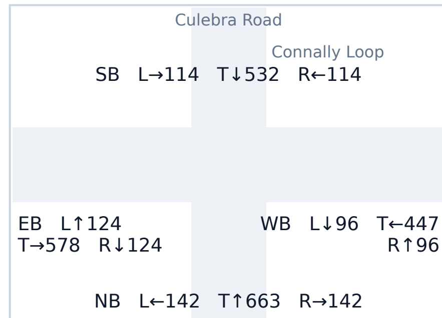
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,178	3	1,181	1.45 → 1.46	235.6 → 237.4	F → F	1,208
SB	944	4	948	1.17 → 1.17	112.3 → 114.1	F → F	969
EB	1,033	0	1,033	1.28 → 1.28	157.9 → 157.9	F → F	1,056
WB	799	0	799	0.99 → 0.99	53.0 → 53.0	D → D	815

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	4	57%
NB Through	3	43%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

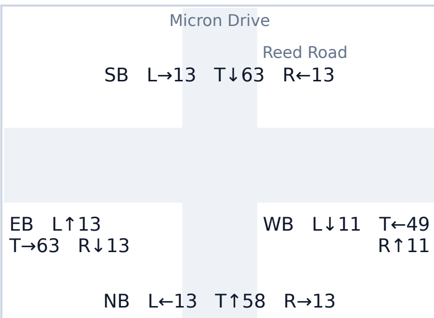
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Micron Drive & Reed Road

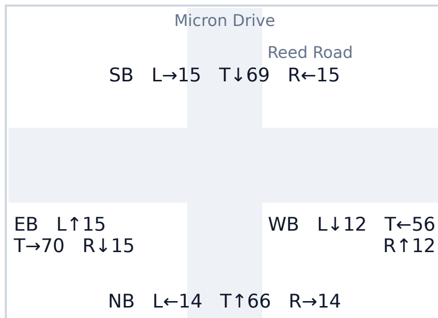
Signal ID	san-antonio-68
Location	SW San Antonio-New Braunfels · 0.84 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS B · 15.6 s/veh · v/c 0.20
Build (PM)	LOS B · 15.6 s/veh · v/c 0.21
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

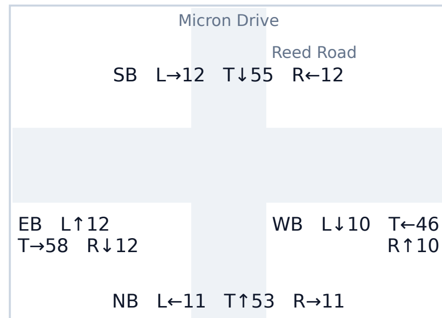
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	93	0	94	0.12 → 0.12	14.6 → 14.6	B → B	56
SB	99	0	99	0.12 → 0.12	14.7 → 14.7	B → B	59
EB	97	3	100	0.12 → 0.12	14.7 → 14.7	B → B	60
WB	77	3	80	0.09 → 0.10	14.5 → 14.5	B → B	48

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	3	50%
EB Through	3	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

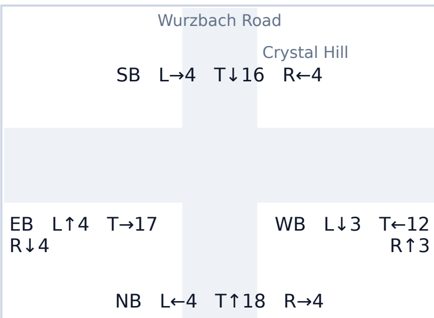
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.16 Wurzbach Road & Crystal Hill

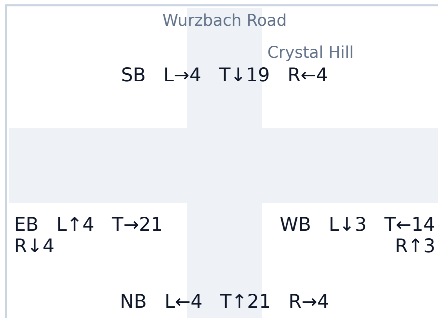
Signal ID	san-antonio-524
Location	SW San Antonio-New Braunfels · 0.94 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

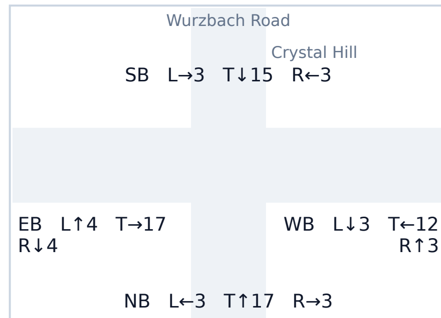
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	29	0	29	0.04 → 0.04	13.9 → 13.9	B → B	17
SB	27	0	27	0.03 → 0.03	13.9 → 13.9	B → B	15
EB	27	2	29	0.03 → 0.04	13.9 → 13.9	B → B	17
WB	19	2	20	0.02 → 0.03	13.8 → 13.8	B → B	12

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	2	50%
WB Through	2	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

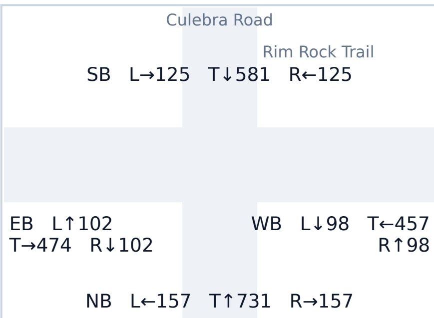
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.17 Culebra Road & Rim Rock Trail

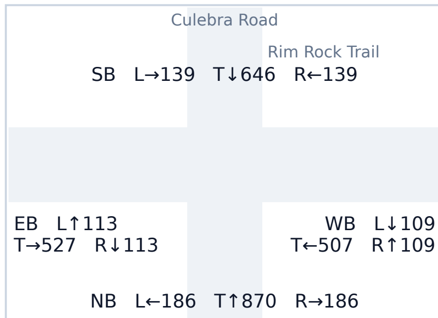
Signal ID	san-antonio-3151
Location	SW San Antonio-New Braunfels · 0.98 mi from site
Added PM peak trips	145
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.94
Build (PM)	LOS F · 300.0 s/veh · v/c 2.02
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)

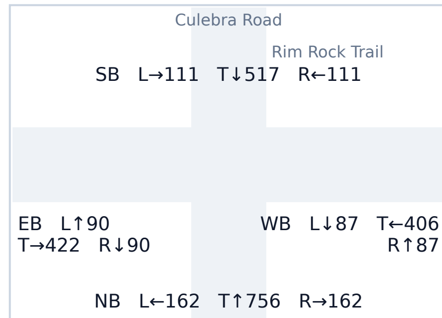
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,097	145	1,242	1.35 → 1.53	192.1 → 270.6	F → F	1,270
SB	924	0	924	1.14 → 1.14	102.3 → 102.3	F → F	945
EB	753	0	753	0.93 → 0.93	41.9 → 41.9	D → D	734
WB	725	0	725	0.90 → 0.90	37.3 → 37.3	D → D	689

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	145	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.